

DSAI 490 - Assignment 1

Assignment: Representation Learning with Autoencoders (AE & VAE)

Assignment Context

Modern machine learning systems rely heavily on the ability to learn **compact and meaningful representations** of data. Autoencoders and their probabilistic extension, Variational Autoencoders (VAEs), are powerful tools for **unsupervised learning, dimensionality reduction, and generative modeling**.

In real-world applications such as medical imaging, anomaly detection, and data compression, understanding how latent representations are formed and utilized is critical.

Your challenge in this assignment is to design and implement **Autoencoder (AE)** and **Variational Autoencoder (VAE)** models, analyze their behavior, and explore their ability to reconstruct and generate data.

Objectives

- Develop and train a **Autoencoder (AE)** for data reconstruction
 - Implement a **Variational Autoencoder (VAE)** with probabilistic latent space
 - Design and visualize **latent representations**
 - Compare AE and VAE in terms of reconstruction and generation
 - Explore applications such as **denoising and sample generation**
-

Expected Features

Feature	Description
AE implementation	Encoder-decoder architecture for data reconstruction
VAE implementation	Probabilistic latent space with reparameterization trick
Latent space visualization	2D/3D visualization of learned representations
Reconstruction analysis	Compare original vs reconstructed outputs
Sample generation	Generate new samples from latent space (VAE)
Denoising capability	Model robustness to noisy inputs
Loss visualization	Track reconstruction and KL divergence losses

Deliverables

- 1. Codebase**
 - Well-structured and modular implementation of AE and VAE
 - Clear separation between models, training, and utilities

(Submit via GitHub repository)

 - *You should use our course **Code Conventions***
 - ***You should use `tf.data`***
 - ***Use the given dataset. Do NOT search for or use an `.npz` or `.csv` version. it must be uploaded to your Google Drive as required.***
 - ***Upload the dataset to your Google Drive, extract it, and use it after mounting Google Drive OR make everything works local on your computer***
- 2. Experiment Notebook / Script**
 - Training pipeline with visualizations
 - Includes reconstruction, latent space plots, and generated samples
- 3. Technical Report (max 2 pages)**
 - Model architectures (AE & VAE)
 - Key differences and observations
 - Latent space behavior analysis
 - Results and insights
- 4. Video Demonstration (Mandatory)**
 - 2–5 minute recording showing:

- Model training / outputs
- Latent space visualization
- Generated samples
- Key findings

Evaluation Criteria

Criteria	Weight	Description
Reconstruction Quality	25%	Accuracy of reconstructed outputs
Latent Space Understanding	20%	Quality of visualization and interpretation
VAE Implementation	20%	Correct use of probabilistic modeling and KL loss
System Design & Code Quality	15%	Modularity, readability, and structure
Experimentation & Insights	10%	Depth of analysis and comparisons
Presentation & Report	10%	Clarity of explanation and communication
